

Question	Answer	Mark	Mark scheme	Additional guidance
17 (a)	Shown	M1	For a start to establish at least one root in $[1,2]$, eg substitutes $x = 1$ or $x = 2$ into LHS, eg $1^3 + 2 \times 1 - 6$, or $2^3 + 2 \times 2 - 6$	$f(x) = x^3 + 2x - 6$ $f(1) = -3$ $f(2) = 6$
		C1	For a complete argument, eg evaluates for both values of x and states since there is a sign change there must be at least one root in $1 < x < 2$ (as f is continuous)	
(b)	Shows re-arrangement	C1	for showing re-arrangement; must see $x^3 + 2x - 6 = 0$ leading to $x(x^2 + 2) - 6 = 0$ or $x(x^2 + 2) = 6$ or $x = \frac{6}{x^2 + 2}$ leading to $x^3 + 2x = 6$	
(c)	1.4496	M1	for $x_1 = \frac{6}{1.45^2 + 2}$ (=1.462(522851...))	If a correct value is given and then rounded or rounded incorrectly award full marks.
		M1	for $x_2 = \frac{6}{(1.462(52...))^2 + 2}$ (=1.449634937)	
		A1	for answer in the range 1.4496 to 1.4497	